

Lovász–Kneser Theorem

The Lovász–Kneser theorem states that the chromatic number of the Kneser graph $KG(n, k)$ is exactly $n - 2k + 2$. Proved by László Lovász in 1978, it resolved Martin Kneser's 1955 combinatorial conjecture and successfully launched the field of topological combinatorics.

Core Concepts

- **Kneser Graph** $KG(n, k)$: Vertices = all $\binom{n}{k}$ k -subsets of $[n] = \{1, \dots, n\}$. Edge between A, B iff $A \cap B = \emptyset$.

$$V(KG(n, k)) = \{A \subset [n] : |A| = k\}, \quad E = \{\{A, B\} : A \cap B = \emptyset\}$$

Interpretation: each vertex is a committee of size k from n people. Edge means committees completely disjoint — no common member. So graph encodes disjointness relation.

Size:

$$|V(KG(n, k))| = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Degree: fix A , number of B disjoint from A is $\binom{n-k}{k}$, so regular degree $\binom{n-k}{k}$ if $n \geq 2k$, otherwise 0.

- **Chromatic Number** $\chi(G)$: Smallest c so vertices can be colored with c colors, no edge monochromatic. Formal:

$$\chi(G) = \min\{c : \exists f : V \rightarrow [c], uv \in E \implies f(u) \neq f(v)\}$$

χ measures how much intersection structure forces diversity. If graph has no edges, $\chi = 1$. If contains clique size ω , then $\chi \geq \omega$.

- **Kneser's Conjecture 1955**: $\chi(KG(n, k)) = n - 2k + 2$ for $n \geq 2k$.

If $n < 2k$, any two k -sets intersect by pigeonhole principle:

$$|A| + |B| = 2k > n \implies |A \cap B| \geq 2k - n \geq 1$$

So $E = \emptyset$, edgeless, $\chi = 1$. Interesting only when $n \geq 2k$, disjoint pairs exist.

Kneser intuition: to color so disjoint sets get different colors, need $n - 2k + 2$ colors. Try with fewer, you must give same color to two disjoint sets — violates coloring.

Deep expansion of each concept

1. Kneser Graph $KG(n, k)$

Ground set:

$$[n] = \{1, 2, \dots, n\}$$

Vertices: Every k -element subset

$$\binom{[n]}{k} = \{A \subset [n] : |A| = k\}$$

Number

$$|V| = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Examples:

$$n = 5, k = 2 : |V| = \binom{5}{2} = 10$$

$$V = \{12, 13, 14, 15, 23, 24, 25, 34, 35, 45\}$$

$$n = 6, k = 2 : |V| = 15$$

$$n = 7, k = 3 : |V| = 35$$

$$n = 10, k = 3 : |V| = 120$$

We abbreviate $\{1, 2\}$ as 12.

Edges: Unordered pair $\{A, B\}$ is edge iff

$$A \cap B = \emptyset$$

No common element. So 12 adjacent to 34 because $\{1, 2\} \cap \{3, 4\} = \emptyset$, but 12 not adjacent to 23 because intersection $\{2\} \neq \emptyset$.

Degree calculation:

Fix A , $|A| = k$. Remaining ground $[n] \setminus A$ size $n - k$. To be disjoint from A , B must be chosen entirely from remaining $n - k$ elements:

$$\deg(A) = \binom{n-k}{k} \quad \text{if } n \geq 2k$$

If $n < 2k$, $n - k < k$, binomial 0, no disjoint k -set, degree 0, graph edgeless.

Thus when $n \geq 2k$, $KG(n, k)$ is regular of degree $\binom{n-k}{k}$.

Example $KG(5, 2)$: $\deg = \binom{3}{2} = 3$, cubic 10-vertex Petersen.

$KG(6, 2)$: $\deg = \binom{4}{2} = 6$, 15 vertices degree 6.

Clique number: Clique = set of vertices pairwise disjoint k -sets. How many pairwise disjoint k -sets can pack into $[n]$? At most $\left\lfloor \frac{n}{k} \right\rfloor$. So

$$\omega(KG(n, k)) = \left\lfloor \frac{n}{k} \right\rfloor$$

For $n = 5, k = 2$, $\omega = 2$ — just an edge, no triangle of pairwise disjoint 2-sets needs 6 distinct elements. For $n = 7, k = 2$, $\omega = 3$: $\{12, 34, 56\}$ pairwise disjoint.

Clique number much smaller than conjectured chromatic number $n - 2k + 2$ when k large. e.g. $n = 2k + 1$: $\omega = 2$, but $n - 2k + 2 = 3$. So $\chi > \omega$.

Interpretations:

- Committees: n people, each committee k people, edge when committees have no common member — conflict graph for sharing member.

- Constant-weight codes: binary strings length n , weight k $\left(\text{\textcolor{red}{ones}}\right)$, $\text{two strings adjacent if supports disjoint} \iff \text{Hamming distance } 2k$. Independent set = intersecting family (pairwise intersect).
- Scheduling: k -sets are resource requirements, disjoint means can run in parallel? Actually edge = conflict? Depends, but disjointness often conflict for coloring.

Symmetry: $KG(n, k)$ vertex-transitive: symmetric group S_n permutes ground set, induces automorphism of graph. Highly symmetric, no distinguished vertex, combinatorial lower bound proofs difficult because no obvious bottleneck.

2. Chromatic Number $\chi(G)$ — What It Measures

$$\chi(G) = \min\{c : \exists f : V \rightarrow [c], uv \in E \implies f(u) \neq f(v)\}$$

Proper coloring: assign color label $f(v) \in \{1, \dots, c\}$ so adjacent vertices different colors. No edge monochromatic.

Intuition: Color classes are independent sets — sets with no internal edges.

$$V = I_1 \sqcup I_2 \sqcup \dots \sqcup I_c, \quad I_i \text{ independent}$$

For $KG(n, k)$, independent set = family of k -sets pairwise intersecting (not necessarily all share one common element, but no two disjoint). Erdős–Ko–Rado says for $n \geq 2k$, largest intersecting family size $\binom{n-1}{k-1}$ — all sets containing fixed element 1.

Bounds:

$$\omega(G) \leq \chi(G) \leq \Delta(G) + 1$$

ω clique needs distinct colors for each vertex in clique. Δ max degree greedy gives $\Delta + 1$ colors suffice.

For Kneser: $\omega = \lfloor n/k \rfloor$, $\Delta = \binom{n-k}{k}$ huge, so $\Delta + 1$ bound useless, gives $\chi \leq \binom{n-k}{k} + 1$ far larger than $n - 2k + 2$.

If graph edgeless, $\chi = 1$ — one color paints all vertices, no conflict.

Why χ matters: Minimum number of groups to partition vertices so no conflict inside group. For Kneser, minimum number of intersecting families to partition all k -sets.

Example calculations:

$KG(5, 2)$: 10 vertices, cubic, contains odd cycle, $\chi = 3$

$KG(6, 2)$: 15 vertices, degree 6, $\chi = 6 - 4 + 2 = 4$

$KG(7, 2)$: $\chi = 7 - 4 + 2 = 5$

$KG(7, 3)$: $\binom{7}{3} = 35$ vertices, degree $\binom{4}{3} = 4$, $\chi = 7 - 6 + 2 = 3$

Notice $KG(7, 3)$ has 35 vertices degree 4, yet needs 3 colors — sparse but chromatic 3.

3. Kneser's Conjecture 1955 — Intuition

Statement:

$$\boxed{\chi(KG(n, k)) = n - 2k + 2 \quad \text{for } n \geq 2k}$$

If $n < 2k$, $\chi = 1$ trivial by pigeonhole.

Proof of trivial case $n < 2k$:

Let A, B be any two k -sets. $|A \cup B| = |A| + |B| - |A \cap B| = 2k - |A \cap B| \leq n$. So

$$|A \cap B| \geq 2k - n$$

If $n < 2k$, then $2k - n \geq 1$, so $|A \cap B| \geq 1$, always intersect. Hence no edge:

$$E(KG(n, k)) = \emptyset \implies \chi = 1$$

Why conjecture plausible — upper bound easy, lower bound hard:

Upper bound $\chi \leq n - 2k + 2$ via anchor coloring (see general strategy):

For $i = 1$ to $n - 2k + 1$: color all k -sets containing i and not containing $1, \dots, i - 1$ with color i

Final color $n - 2k + 2$: all k -sets $\subseteq \{n - 2k + 2, \dots, n\}$, size $2k - 1$ pool

Final pool size $2k - 1$, any two k -sets intersect, so independent.

Thus $n - 2k + 2$ colors suffice — easy explicit construction.

Lower bound $\chi \geq n - 2k + 2$ says cannot do with $n - 2k + 1$ colors. Try to color with fewer, you'd be forced to give same color to two disjoint sets. Why forced? Not obvious combinatorially. For $KG(6, 2)$, try 3 colors for 15 vertices degree 6 — seems maybe possible? But formula says need 4. Need deep reason.

Intuition for necessity: Suppose we have $m = n - 2k + 1$ color classes, each independent = intersecting family. Need to cover all $\binom{n}{k}$ sets by m intersecting families. Is it possible? Erdős–Ko–Rado says largest intersecting families are stars $\mathcal{F}_i = \{A : i \in A\}$. Stars centered at $1, \dots, m$ cover many sets, but sets completely contained in complement $\{m + 1, \dots, n\}$ of size $2k - 1$ avoid all those centers. That complement contains $\binom{2k - 1}{k}$ sets, pairwise intersecting, could be one more family, but that would be $m + 1$ -th color. So with only m stars you cannot cover all. But intersecting families need not be stars — Hilton–Milner shows non-trivial intersecting families exist — so need topology to rule out all possibilities.

Example $n = 6, k = 2, m = 3$ attempt fails: Need 3 intersecting families cover all 15 edges-sets. Stars at 1,2,3 cover: star1 = $\{12,13,14,15,16\}$ 5 sets, star2 = $\{23,24,25,26\}$ 4 new, star3 = $\{34,35,36\}$ 3 new, total 12, missing $\{45,46,56\}$ 3 sets — they intersect each other but need 4th color. Any other intersecting families also fail — proved only via Borsuk–Ulam.

Dimension intuition: $n - 2k + 2$ grows with n linearly, while $\omega = \lfloor n/k \rfloor$ grows slower when k large. So chromatic number much larger than clique number — Kneser graphs are not perfect, have high χ despite small cliques. This makes them interesting counterexamples and test cases.

In short: $KG(n, k)$ encodes disjointness, $\binom{n}{k}$ vertices, degree $\binom{n - k}{k}$, clique number $\lfloor n/k \rfloor$, chromatic number measures partition into intersecting families, trivial $\chi = 1$ when $n < 2k$ by $|A \cap B| \geq 2k - n$, and Kneser conjectured exact value $n - 2k + 2$ for $n \geq 2k$ — upper bound easy by $n - 2k + 1$ stars plus final $2k - 1$ pool, lower bound deep requiring Borsuk–Ulam, resolved by Lovász 1978 launching topological combinatorics.

History, Discovery, Modern Developments

Kneser 1955 — Erdős — Lovász 1978 — Bárány 1978 — Schrijver 1978 — Dol'nikov-Kriz — Matoušek 2004 — Modern

Martin Kneser (1928–2004), German algebraist, 1955: In *Aufgabe 360* in *Jahresbericht der Deutschen Mathematiker-Vereinigung*, observed easy upper bound $\chi(KG(n, k)) \leq n - 2k + 2$ via explicit coloring (anchor colors, see below), and

conjectured equality. Could not prove lower bound. Problem looked purely combinatorial.

1970s — gap: For 23 years attempts failed. Paul Erdős popularized problem, asked about chromatic number, offered prizes. Combinatorial methods failed because Kneser graphs highly symmetric, vertex-transitive, no obvious small bottleneck. Standard bounds like clique number $\omega(KG(n, k)) = \left\lfloor \frac{n}{k} \right\rfloor$ (partition $[n]$ into disjoint k -sets) gives only $\chi \geq \lfloor n/k \rfloor$, far from $n - 2k + 2$ when k large. e.g. $n = 2k + 1$: $\lfloor n/k \rfloor = 2$ but conjectured $\chi = 3$.

László Lovász (1948–), 1978: Then 29, in Szeged, Hungary. Proved lower bound using **Borsuk–Ulam theorem**. Paper "*Kneser's conjecture, chromatic number, and homotopy*" in *J. Combinatorial Theory A*.

Introduced **neighborhood complex** $N(G)$: simplicial complex where vertices are vertices of G , simplex $\sigma = \{v_1, \dots, v_r\}$ is simplex iff σ has common neighbor: $\exists w : \{w, v_i\} \in E \ \forall i$. So $N(G)$ encodes overlapping neighborhoods.

Lovász showed: if G is m -colorable, then $N(G)$ is $(m - 1)$ -connected? Actually: if $N(G)$ is k -connected then $\chi(G) \geq k + 3$. More precisely, connectivity of $N(G)$ lower bounds chromatic number.

For $KG(n, k)$, he computed homotopy type: $N(KG(n, k))$ homotopy equivalent to sphere S^{n-2k} ? Actually wedge of spheres, but at least $(n - 2k - 1)$ -connected. Using Borsuk–Ulam to show non-trivial homology, deduced $\chi \geq n - 2k + 2$.

This was first major use of algebraic topology to solve discrete problem — birth of **topological combinatorics**.

Imre Bárány 1978: Shortly after Lovász, gave even shorter proof using same Borsuk–Ulam but more direct, avoiding neighborhood complex language, using Gale's lemma about placing points on sphere.

Alexander Schrijver 1978: Found subgraph $SG(n, k)$ (stable Kneser / Schrijver graph): vertices are k -sets with no two cyclic consecutive elements in $[n]$ arranged on circle, and no pair containing both 1 and n . Still has $\chi = n - 2k + 2$ but vertex-critical — removing any vertex drops chromatic number. Shows Lovász bound tight for minimal graphs. $|V(SG(n, k))|$ much smaller than $\binom{n}{k}$. For example $SG(5, 2) = C_5$, 5-cycle, $\chi = 3$.

1980s–2000s — is topology necessary? Question: is there purely combinatorial proof of Kneser? Jiří Matoušek 2004 gave proof that still uses Borsuk–Ulam but avoids simplicial complexes, making it accessible to combinatorialists. Uses Lemma: there exists set $X \subset S^d$ of n points in general position such that every open hemisphere contains at least k points? Actually Gale's lemma: n points on S^{n-2k} can be placed so every open hemisphere contains at least k points. This is key.

Borsuk–Ulam: If $f : S^d \rightarrow \mathbb{R}^d$ continuous, $\exists x : f(x) = f(-x)$

Equivalent **Lyusternik–Schnirelmann**: If S^d covered by $d + 1$ closed sets, one contains antipodal pair $x, -x$. Or open sets.

This equivalence used to derive contradiction.

2000s–now — modern developments:

- **Dol'nikov 1988, Kriz 1992, 2000:** Generalization: For family $\mathcal{F}$ of subsets, define Kneser graph $KG(\mathcal{F})$ where vertices are sets in $\mathcal{F}$, edges disjoint. Define 2-colorable defect $\text{cd}_2(\mathcal{F}) = \text{minimum size of set to delete from ground set so remaining family 2-colorable (no monochromatic set)}$. Then $\chi(KG(\mathcal{F})) \geq \text{cd}_2(\mathcal{F})$. Lovász–Kneser is special case where $\mathcal{F} = \binom{[n]}{k}$, $\text{cd}_2 = n - 2k + 2$. Proof uses $\mathbb{Z}_2$ -index $\text{ind}_{\mathbb{Z}_2}$.
- **Ziegler, Björner, de Longueville:** Systematic theory of $\mathbb{Z}_2$ -spaces, box complexes $B(G)$, Hom-complexes $\text{Hom}(G, H)$. Box complex $B(G)$: vertices $V(G) \times \{0, 1\}$, etc., its $\mathbb{Z}_2$ -connectivity gives lower bound $\chi(G) \geq \text{ind}(B(G)) + 2$.

- **Alishahi–Hajjabolhassan:** Chromatic number of Kneser hypergraphs where edge = r pairwise disjoint sets, χ of Kneser hypergraph $KG^r(n, k)$.
- **Computational:** $KG(n, k)$ huge $\binom{n}{k}$ vertices, e.g. $KG(100, 40)$ astronomically huge, but its chromatic number known exactly $100 - 80 + 2 = 22$ — rare case where χ computable in closed form, while computing χ generally NP-hard.

Modern day uses:

Topological lower bounds used for scheduling, coding, fair division

- **Combinatorial geometry:** Proof method used for ham sandwich $S^d \rightarrow \mathbb{R}^d$, necklace splitting k thieves fairly split necklace with t types using $\leq t(k-1)$ cuts, Tverberg theorem $(d+1)(r-1)+1$ points in $\mathbb{R}^d$ can be partitioned into r parts whose convex hulls intersect, consensus halving.
- **Complexity:** Kneser graphs are test cases for hardness of graph coloring approximation. $\chi(KG(n, k))$ known, so can test algorithms that try to color with fewer colors — they must fail.
- **Coding theory:** Kneser graphs appear as conflict graphs for constant-weight codes: vertices = weight- k binary strings length n (characteristic vectors of k -sets), edge = disjoint support $\implies$ Hamming distance $2k$. Independent set = intersecting family, max size given by Erdős–Ko–Rado $\binom{n-1}{k-1}$. Chromatic number = partition of all k -sets into intersecting families.
- **Equivariant topology:** Lovász's neighborhood complex led to Hom-complexes $\text{Hom}(G, H)$ used to bound $\chi(G)$ via $\mathbb{Z}_2$ -index. $\text{Hom}(K_2, G)$ is essentially neighborhood complex.

Topological Proof — Detailed in Steps

Goal: show $\chi(KG(n, k)) \geq n - 2k + 2$. Upper bound easy (anchor coloring). Lower bound hard — need to prove fewer colors impossible.

1. Topological Foundation — Borsuk–Ulam

$$\boxed{\text{If } f : S^d \rightarrow \mathbb{R}^d \text{ continuous, } \exists x \in S^d : f(x) = f(-x)}$$

What is S^d ?

$$S^d = \{x \in \mathbb{R}^{d+1} : \|x\| = 1\} = \{x : x_1^2 + \cdots + x_{d+1}^2 = 1\}$$

- $S^0 = \{-1, 1\}$ two points
- $S^1 = \{x_1^2 + x_2^2 = 1\}$ circle
- $S^2 = \{x_1^2 + x_2^2 + x_3^2 = 1\}$ ordinary sphere surface
- S^d d -dimensional sphere

Antipodal map $A : S^d \rightarrow S^d$, $A(x) = -x$, opposite point, $A(A(x)) = x$, free involution no fixed point.

Picture:

- $d = 1$: S^1 circle to $\mathbb{R}$ line, continuous $f : S^1 \rightarrow \mathbb{R}$ temperature around circle, opposite points same height — Intermediate Value Theorem forces $\exists x : f(x) = f(-x)$. Indeed define $g(x) = f(x) - f(-x)$, $g(-x) = -g(x)$ odd, if $g(x_0) > 0$ then $g(-x_0) < 0$, IVT gives zero.

- $d = 2$: Earth S^2 to plane $\mathbb{R}^2$ (temperature, pressure) $\rightarrow$ antipodal points same temperature and pressure simultaneously.
Not just temperature equal somewhere and pressure equal elsewhere, same pair for both.

Equivalent LS formulation: Cover S^d by $d + 1$ open sets $U_1, \dots, U_{d+1}$, one contains antipodal pair $x, -x$. Closed version same: If $S^d = F_1 \cup \dots \cup F_{d+1}$ closed, some F_i contains $x, -x$.

Why equivalent? Given $f : S^d \rightarrow \mathbb{R}^d$, suppose no x with $f(x) = f(-x)$, then $g(x) = f(x) - f(-x) \neq 0$, define $h(x) = g(x)/\|g(x)\| : S^d \rightarrow S^{d-1}$ odd. Then sets $U_i^+ = \{x : h_i(x) > 0\}$, $U_i^- = \{x : h_i(x) < 0\}$ cover, no contains antipodal pair — contradict LS.

This LS version directly used for Kneser: if we could color with d colors, we would get covering of S^d by d open sets none containing antipodal pair, plus one more set for equator — total $d + 1$ sets, LS says one contains antipodal pair, contradiction.

Why Borsuk–Ulam deep: Degree theory. Odd map $S^d \rightarrow S^d$ has odd degree $\neq 0$, cannot factor through S^{d-1} which has degree 0. Homology $H_d(S^d) \cong \mathbb{Z}$, $H_d(S^{d-1}) = 0$.

2. Place points on sphere in general position

Ground set $[n] = \{1, \dots, n\}$. Let

$$d = n - 2k + 1, \quad m = d$$

We work on sphere $S^d \subset \mathbb{R}^{d+1}$. Place n points $v_1, \dots, v_n \in S^d$ in general position:

No $d + 1$ points lie in a hyperplane through origin

That is, no $d + 1$ points lie on a great S^{d-1} equator.

How to achieve? Moment curve trick: $\gamma(t) = (1, t, t^2, \dots, t^d) \in \mathbb{R}^{d+1}$, normalize $v(t) = \gamma(t)/\|\gamma(t)\| \in S^d$. Any $d + 1$ distinct t give linearly independent vectors $\gamma(t)$ (Vandermonde determinant $\neq 0$), so no hyperplane through origin contains $d + 1$ of them.

Gale's lemma strengthened: For $n \geq 2k$, exists set $X \subset S^{n-2k}$ of n points such that every open hemisphere contains at least k points. Proof uses same moment curve and ham sandwich. We need one dimension higher $S^{n-2k+1} = S^d$, so we can also ensure general position plus property that any open hemisphere contains at least k points? Actually in Matoušek proof we place on S^d with $d = n - 2k + 1$, then property: every open hemisphere contains at least k points? Check: For $d = n - 2k$, every open hemisphere contains at least k points. For $d = n - 2k + 1$, every open hemisphere contains at least $k + 1$? Let's not confuse, we only need general position for counting argument later. The key property for contradiction is that if both open hemispheres $H(x^*), H(-x^*)$ contain $< k$ points, then $\geq n - 2k + 2$ points on equator, contradicting general position bound $\leq d = n - 2k + 1$.

Why important: ensures that if open hemisphere $H(x)$ contains $< k$ points of $[n]$, then complement contains many points. For our contradiction, we will deduce both $H(x^*), H(-x^*)$ contain $< k$ points, forcing many points on equator.

Visualization: n points sprinkled on sphere fairly evenly, no cluster on equator. Any great circle $E(x)$ can contain at most d points.

3. Proof by contradiction — assume too few colors

Suppose $KG(n, k)$ colorable with $m = n - 2k + 1$ colors $\{1, \dots, m\}$. Want contradiction.

Let

$$c : \binom{[n]}{k} \rightarrow [m]$$

proper coloring:

$$A \cap B = \emptyset \implies c(A) \neq c(B)$$

So each color class

$$\mathcal{F}_i = c^{-1}(i) = \{A : c(A) = i\}$$

is intersecting family: any two sets in same $\mathcal{F}_i$ intersect (otherwise edge monochromatic).

We have m intersecting families covering all $\binom{[n]}{k}$ k -sets:

$$\binom{[n]}{k} = \bigcup_{i=1}^m \mathcal{F}_i$$

Goal show impossible when $m = n - 2k + 1$.

4. Define continuous map $f : S^d \rightarrow \mathbb{R}^d$ construction

For $x \in S^d$, define open hemisphere centered at x :

$$H(x) = \{y \in S^d : \langle x, y \rangle > 0\}$$

$\langle \cdot, \cdot \rangle$ standard inner product. $H(x)$ all points making acute angle $< 90^\circ$ with x .

Properties:

$$\begin{aligned} H(-x) &= \{y : \langle -x, y \rangle > 0\} = \{y : \langle x, y \rangle < 0\} = -H(x) \\ H(x) \cap H(-x) &= \emptyset \\ E(x) &= \{y : \langle x, y \rangle = 0\} = S^{d-1} \text{ equator separates them} \\ S^d &= H(x) \sqcup E(x) \sqcup H(-x) \end{aligned}$$

For a k -set $F \subset [n]$, say $F \subset H(x)$ means all its points lie in hemisphere:

$$F \subset H(x) \iff \forall j \in F : v_j \in H(x) \iff \forall j \in F : \langle x, v_j \rangle > 0$$

Set

$$H_F = \{x \in S^d : F \subset H(x)\} = \bigcap_{j \in F} \{x : \langle x, v_j \rangle > 0\}$$

Intersection of k open hemispheres $\{x : \langle x, v_j \rangle > 0\}$, each open, so H_F open
(maybe empty if points not contained in any open hemisphere, but Gale ensures many non-empty).

Now define for each color i :

$$U_i = \bigcup_{F: c(F)=i} H_F$$

U_i open as union of open sets. $x \in U_i$ iff $H(x)$ contains at least one k -set colored i .

Idea: $f_i(x) > 0$ iff $x \in U_i$.

Matoušek explicit bump functions:

For each F , define continuous function $\phi_F : S^d \rightarrow [0, 1]$ supported in H_F , e.g.

$$\phi_F(x) = \max \left\{ 0, \min_{j \in F} \langle x, v_j \rangle \right\}$$

$\phi_F(x) > 0 \iff F \subset H(x)$ and distance from boundary positive. Then define

$$f_i(x) = \sum_{F: c(F)=i} \phi_F(x)$$

Sum over all F colored i . Since each ϕ_F continuous, finite sum of continuous? Actually infinite sum but $\binom{n}{k}$ finite, so finite sum, continuous. Moreover $f_i(x) > 0 \iff \exists F : c(F) = i, F \subset H(x)$.

Alternative definition using distance to complement:

$$f_i(x) = \begin{cases} \text{dist}(x, S^d \setminus U_i) & x \in U_i \\ 0 & x \notin U_i \end{cases}$$

Distance to closed complement continuous, zero outside U_i , positive inside.

Thus

$$f(x) = (f_1(x), \dots, f_d(x)) \in \mathbb{R}^d \text{ continuous}$$

$d = m = n - 2k + 1$ coordinates matching colors.

Intuition: f measures, for each color, how deeply hemisphere $H(x)$ contains a k -set of that color. If hemisphere contains none, coordinate zero.

5. Apply Borsuk–Ulam

f continuous $S^d \rightarrow \mathbb{R}^d$, where $d = n - 2k + 1$.

Borsuk–Ulam gives $\exists x^* \in S^d$ with

$$f(x^*) = f(-x^*)$$

That is for each i ,

$$f_i(x^*) = f_i(-x^*)$$

6. Expose combinatorial contradiction

$H(x^*)$ and $H(-x^*)$ disjoint opposite hemispheres.

Lemma: For each color i , at most one of $H(x^*)$, $H(-x^*)$ can contain a k -set colored i .

Proof: Suppose $H(x^*)$ contains A colored i and $H(-x^*)$ contains B colored i . Then

$$A \subset H(x^*), B \subset H(-x^*), A \cap B = \emptyset$$

Why disjoint? If $j \in A$, then $v_j \in H(x^*)$, so $\langle x^*, v_j \rangle > 0$. If $j \in B$, then $v_j \in H(-x^*)$, so $\langle -x^*, v_j \rangle > 0 \iff \langle x^*, v_j \rangle < 0$. Cannot have both > 0 and < 0 for same j , so j cannot belong to both A and B . Hence $A \cap B = \emptyset$. Then A, B adjacent in $KG(n, k)$ but same color i — violates proper coloring. Contradiction. So at most one hemisphere contains color i .

Thus for each i , not both $f_i(x^*) > 0$ and $f_i(-x^*) > 0$. At most one positive.

But Borsuk–Ulam equality $f_i(x^*) = f_i(-x^*)$ says they are equal. If one positive, both positive — impossible. Therefore

$$f_i(x^*) = 0 \quad \forall i = 1, \dots, d$$

Meaning: neither $H(x^*)$ nor $H(-x^*)$ contains any k -set colored with colors $1, \dots, d$.

Since we assumed all $\binom{n}{k}$ vertices colored with these d colors, any k -set contained in $H(x^*)$ would need a color, contradiction unless $H(x^*)$ contains no k -set at all. Similarly for $H(-x^*)$.

When does open hemisphere $H(x^*)$ contain no k -set? Exactly when number of ground points inside $H(x^*)$ is $< k$:

$$|\{j : v_j \in H(x^*)\}| \leq k - 1$$

Because if $\geq k$ points lie in $H(x^*)$, then those k points form a k -set contained in $H(x^*)$, which would need a color. So must be $\leq k - 1$.

Similarly

$$|\{j : v_j \in H(-x^*)\}| \leq k - 1$$

Total points in both open hemispheres:

$$|H(x^*)| + |H(-x^*)| \leq 2k - 2$$

Remaining points lie on equator $E(x^*) = S^{d-1} = \{y : \langle x^*, y \rangle = 0\}$:

$$|\{j : v_j \in E(x^*)\}| = n - |H(x^*)| - |H(-x^*)| \geq n - (2k - 2) = n - 2k + 2$$

But equator is intersection of S^d with hyperplane through origin orthogonal to x^* : $\{y : \langle x^*, y \rangle = 0\}$. By general position assumption, no $d + 1 = n - 2k + 2$ points lie in a hyperplane through origin. At most $d = n - 2k + 1$ points can lie on equator.

We have $\geq n - 2k + 2$ points on equator, contradiction because

$$n - 2k + 2 > n - 2k + 1 = d$$

Thus assumption $m = n - 2k + 1$ colorable false.

Hence need at least $m + 1 = n - 2k + 2$ colors.

Combined with upper bound anchor coloring giving $\chi \leq n - 2k + 2$:

$$\begin{aligned} \chi(KG(n, k)) &\geq n - 2k + 2 \\ \chi(KG(n, k)) &\leq n - 2k + 2 \\ \implies \chi(KG(n, k)) &= n - 2k + 2 \end{aligned}$$

QED.

Remarks on original Lovász proof: Used neighborhood complex $N(KG(n, k))$ and showed its connectivity $\geq n - 2k - 1$, then $\chi \geq \text{conn} + 3$. Matoušek version above avoids simplicial complexes, uses only Borsuk–Ulam and Gale's lemma, more elementary but same topological core.

This completes lower bound, establishing Lovász–Kneser theorem.

The General Coloring Strategy — Upper Bound

Why $n - 2k + 2$ colors sufficient? Constructive greedy anchor. This is the easy half of Lovász–Kneser, giving

$$\chi(KG(n, k)) \leq n - 2k + 2$$

Combined with hard topological lower bound $\geq n - 2k + 2$, gives equality.

Setup

Let

$$c = n - 2k + 2$$

We will color all $\binom{n}{k}$ vertices with c colors.

Colors $1, \dots, c - 1 = n - 2k + 1$ colors = anchors

Color c = final dump = leftover pool

Ground set

$$[n] = \{1, 2, \dots, n\}$$

Split ground into two parts:

$$\begin{aligned} A &= \{1, 2, \dots, n - 2k + 1\}, & |A| &= n - 2k + 1 = c - 1 \\ B &= \{n - 2k + 2, \dots, n\}, & |B| &= n - (n - 2k + 1) = 2k - 1 \end{aligned}$$

Note $[A] \sqcup B = [n]$, $|A| + |B| = n$, $|B| = 2k - 1$ critical size.

Procedure — Greedy Anchor

For each k -set $F \subset [n]$:

For $i = 1$ to $n - 2k + 1$:

if $i \in F$ then $\text{color}(F) = i$ and stop

If loop ends, $F \subseteq \{n - 2k + 2, \dots, n\} = B$

$\text{color}(F) = c$

In words: scan anchors $1, 2, \dots, n - 2k + 1$ in order, assign first anchor that belongs to F . If F contains none of them, then F lies entirely in B , assign final color c .

This is well-defined because every F either meets A or is contained in B (if $F \cap A = \emptyset \implies F \subseteq [n] \setminus A = B$).

Formally:

$$\text{color}(F) = \begin{cases} \min(F \cap A) & \text{if } F \cap A \neq \emptyset \\ c & \text{if } F \cap A = \emptyset \end{cases}$$

$\min$ uses natural order $1 < \dots < n$.

Validity Detailed — Why Proper?

Need each color class independent: no edge inside class, i.e., no two vertices same color are disjoint.

- **Color $i < n - 2k + 2$:** Suppose $\text{color}(F_1) = i$ and $\text{color}(F_2) = i$, $1 \leq i \leq n - 2k + 1$.

By construction, $i \in F_1$ and $i \in F_2$, and for all $j < i$, $j \notin F_1, F_2$ (otherwise earlier anchor would have been chosen), but crucial is both contain i .

Then

$$F_1 \cap F_2 \supseteq \{i\} \neq \emptyset$$

So F_1, F_2 not disjoint, no edge between them in $KG(n, k)$. Hence color class i is independent set — actually a star centered at i , the classic Erdős–Ko–Rado extremal intersecting family.

Indeed $\mathcal{F}_i = \{F : i \in F, F \cap \{1, \dots, i-1\} = \emptyset\}$ is subfamily of star, but any two contain i , intersect.

- **Final color c :** $\text{color}(F) = c$ means $F \cap A = \emptyset$, so $F \subseteq B$, where $|B| = 2k - 1$.

Let A, B be two k -subsets of B (using A, B for sets now, not ground partition):

$$|A| = |B| = k, A, B \subseteq B, |B| = 2k - 1$$

By pigeonhole principle:

$$|A \cup B| = |A| + |B| - |A \cap B| \leq |B| = 2k - 1$$

So

$$k + k - |A \cap B| \leq 2k - 1 \implies |A \cap B| \geq 2k - (2k - 1) = 1$$

Thus

$$|A \cap B| \geq 1$$

Any two k -sets inside $(2k - 1)$ -set intersect. No disjoint pair, no edge. So final color class independent as well.

Hence coloring proper with $c = n - 2k + 2$ colors.

$$\chi(KG(n, k)) \leq n - 2k + 2$$

Examples — Step by Step

$KG(6, 2): n = 6, k = 2 \implies c = 6 - 4 + 2 = 4$ colors, ground $\{1, \dots, 6\}$, $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$, $|B| = 3 = 2k - 1$.

List all $\binom{6}{2} = 15$ vertices:

Color1 $i = 1 : \{12, 13, 14, 15, 16\}$	contains 1
$12 \cap 13 = \{1\}, 12 \cap 14 = \{1\}, \dots$ all intersect at 1	
Color2 $i = 2 : \{23, 24, 25, 26\}$	contains 2 not 1
$23 = \{2, 3\}, 24 = \{2, 4\}, 23 \cap 24 = \{2\} \neq \emptyset$	
Color3 $i = 3 : \{34, 35, 36\}$	contains 3 not 1,2
$34 \cap 35 = \{3\}$, etc.	
Color4 $c = 4 : \{45, 46, 56\}$	$\subseteq \{4, 5, 6\} = B$
$45 = \{4, 5\}, 46 = \{4, 6\}, 45 \cap 46 = \{4\}$	
$45 \cap 56 = \{5\}, 46 \cap 56 = \{6\}$	

Check no edges inside classes: 45, 46 share 4, not disjoint. So 4-colorable. Lower bound says cannot do with 3.

Count: $5 + 4 + 3 + 3 = 15$ all vertices covered.

$KG(5, 2)$ **Petersen**: $n = 5, k = 2 \implies c = 3, A = \{1, 2\}, B = \{3, 4, 5\}, |B| = 3$.

Color1 : $\{12, 13, 14, 15\}$ contains 1

Color2 : $\{23, 24, 25\}$ contains 2 not 1

Color3 : $\{34, 35, 45\} \subseteq \{3, 4, 5\}$

$4 + 3 + 3 = 10$ vertices, proper 3-coloring. No 2-coloring because contains odd cycle.

$KG(7, 3)$: $n = 7, k = 3 \implies c = 7 - 6 + 2 = 3$ colors. $A = \{1, 2\}, B = \{3, 4, 5, 6, 7\}, |B| = 5 = 2k - 1$. Colors:

Color1 : all 3-sets containing 1, $\binom{6}{2} = 20$ sets but some also contain 1? Actually count

$$|\{F : 1 \in F\}| = \binom{6}{2} = 15$$

Color2 : sets containing 2 not 1, $|\{F : 2 \in F, 1 \notin F\}| = \binom{5}{2} = 10$

Color3 : sets $\subseteq \{3, 4, 5, 6, 7\}, \binom{5}{3} = 10$

$15 + 10 + 10 = 35 = \binom{7}{3}$. Color3: any two 3-sets inside 5-set intersect? $3 + 3 = 6 > 5$, so by pigeonhole intersect at least 1. So independent.

General counting check: Number colored with final color $= \binom{2k-1}{k}$. Number colored with anchor i :

$$|\mathcal{F}_i| = \binom{n-i}{k-1} \text{ with first } i-1 \text{ excluded?}$$

Actually $|\{F : i \in F, 1, \dots, i-1 \notin F\}| = \binom{n-i}{k-1}$. Sum:

$$\sum_{i=1}^{n-2k+1} \binom{n-i}{k-1} + \binom{2k-1}{k} = \binom{n}{k}$$

Identity holds by hockey-stick.

Why $2k - 1$ Is Magic Number

$2k - 1$ is largest size of ground set where every two k -sets intersect. If ground size $2k$, then disjoint pair exists: $\{1, \dots, k\}$ and $\{k+1, \dots, 2k\}$ disjoint. So $2k - 1$ threshold where Kneser graph on that ground becomes edgeless:

$$KG(2k-1, k) \text{ has } \chi = 1$$

Indeed $n = 2k - 1 < 2k$, no edges. So final dump color class is exactly $KG(2k-1, k)$, edgeless, can be one color.

If we tried to use $n - 2k + 1$ colors only (no final dump), leftover ground $2k$ would contain disjoint pair needing extra color. So $n - 2k + 2$ minimal for this anchor scheme.

Optimality and Hilton–Milner

Anchor stars $\mathcal{F}_i = \{F : i \in F\}$ are maximum intersecting families by Erdős–Ko–Rado for $n \geq 2k$. This coloring uses largest possible independent sets, greedy optimal. Hilton–Milner shows non-trivial intersecting families smaller, so anchor optimal for upper bound.

Thus easy upper bound construction gives $\chi \leq n - 2k + 2$, and Lovász topological lower bound gives $\chi \geq n - 2k + 2$, together equality.

In short: scan $1, \dots, n - 2k + 1$ as anchors, color k -set by first anchor it contains, otherwise dump into final pool B of size $2k - 1$ where any two k -sets intersect by pigeonhole $|A \cap B| \geq 1$, so each color class independent, $n - 2k + 2$ colors suffice.

Petersen Graph $KG(5, 2)$

Ground $\{1, 2, 3, 4, 5\}$, vertices $\binom{5}{2} = 10$. List all 10:

12, 13, 14, 15, 23, 24, 25, 34, 35, 45

Edge when disjoint: 12 adjacent to 34, 35, 45 (3 neighbors), etc. Degree $\binom{3}{2} = 3$, cubic graph.

$$\chi(KG(5, 2)) = 5 - 4 + 2 = 3$$

Coloring with $c = 3$:

Color1 : $\{12, 13, 14, 15\}$ share 1 — star centered 1
 Color2 : $\{23, 24, 25\}$ share 2 — remaining containing 2
 Color3 : $\{34, 35, 45\} \subseteq \{3, 4, 5\}$, $|\{3, 4, 5\}| = 3 = 2k - 1$

Valid: $34 \cap 35 = \{3\}$, $34 \cap 45 = \{4\}$, $35 \cap 45 = \{5\}$, no edges.

No 2-coloring possible because $KG(5, 2)$ contains odd cycle e.g. $12 - 34 - 25 - 13 - 45 - 12$ length 5, and odd cycle needs 3 colors. So $\chi = 3$ matches formula.

Petersen is also $SG(5, 2)$ stable Kneser — minimal 3-chromatic.

Why It Matters Today

- **Launch of topological combinatorics:** First proof that continuity / antipodal maps can prove discrete lower bound. Now standard toolkit for $\chi(G)$, e.g., Borsuk graphs $BG(n)$ where vertices points on sphere, edges almost antipodal, χ grows with dimension via Borsuk–Ulam. Schrijver graphs $SG(n, k)$ vertex-critical with same chromatic number.
- **Method paradigm:** Transform coloring problem into $\mathbb{Z}_2$ -equivariant map $S^d \rightarrow \text{complex}$, show map cannot exist via Borsuk–Ulam index. Steps:

$$\text{Graph } G \rightarrow \text{Box complex } B(G) \rightarrow \mathbb{Z}_2\text{-space} \rightarrow \text{index } \text{ind}_{\mathbb{Z}_2} B(G) \rightarrow \chi(G) \geq \text{ind} + 2$$

Used for ham sandwich, necklace splitting n thieves, consensus halving, Tverberg, etc.

- **Sharpness:** Schrijver's $SG(n, k)$ shows bound tight even after deleting many vertices, so topology captures exact obstruction, not just estimate. $SG(n, k)$ still needs $n - 2k + 2$ colors but much sparser — proves coloring difficulty not due to many vertices but topological structure.
- **Pedagogical:** Clean example where easy upper bound $n - 2k + 2$ via greedy, hard lower bound needs topology — illustrates gap between constructing solution and proving optimality. Most combinatorics problems easy to give upper bound, lower bound hard — topology provides powerful lower bound method.
- **Coding and intersecting families:** Erdős–Ko–Rado says largest intersecting family of k -sets size $\binom{n-1}{k-1}$. Partition of all $\binom{n}{k}$ sets into intersecting families needs at least $\frac{\binom{n}{k}}{\binom{n-1}{k-1}} = \frac{n}{k}$ families, but Lovász–Kneser gives stronger $n - 2k + 2$ which for large n about n , larger than n/k when $k > 1$. So chromatic number much larger than simple ratio bound.

In short: $KG(n, k)$ vertices = k -sets, edges = disjointness, $\chi = n - 2k + 2$ by anchor coloring upper bound and Borsuk–Ulam lower bound $S^d \rightarrow \mathbb{R}^d$ collapsing antipodal $x^*, -x^*$, forcing $2k - 1$ pigeonhole violation — Lovász 1978 solving Kneser 1955, spawning topological combinatorics used for fair division, coding, complexity, equivariant topology.